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Hybrid Intelligent Control System for High-Quality Sulfur Production under Uncertainty

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***Abstract*—Modern sulfur production units operate under conditions of considerable variability in feedstock composition, unstable temperature regimes, fluctuations in air flow rate, and hydrogen sulfide content. These factors significantly complicate maintaining the required sulfur quality and reduce the effectiveness of conventional control systems. To address these issues, a hybrid intelligent system is proposed that combines an adaptive neuro-fuzzy inference system, a genetic optimization algorithm, and a Sugeno fuzzy inference module. The proposed architecture simultaneously predicts sulfur quality and searches for optimal control actions while accounting for uncertainty in the input data. To evaluate the effectiveness of the developed approach, a computational experiment was conducted using a set of process data reflecting the actual operating ranges of sulfur production units. The ANFIS model was trained using a hybrid optimization method combining the least-squares algorithm and gradient-based tuning of membership-function parameters. The genetic algorithm was used for multiobjective process optimization aimed at maximizing sulfur yield, increasing the sulfur recovery degree, and minimizing sulfur dioxide concentration in off-gases. The simulation results show that the proposed hybrid intelligent system improves the prediction accuracy of output parameters, reduces the model mean squared error, is robust to uncertainty in the input data, and enables automatic selection of optimal unit operating modes.**

Keywords—hybrid intelligent system; ANFIS; genetic algorithm; fuzzy logic; intelligent control.

I. INTDUCTION

The production of high-quality sulfur used in medicine, the chemical and metallurgical industries, and other sectors is one of the most important processes at oil refineries. Stricter environmental requirements for sulfur-compound emissions, higher sulfur-quality requirements, and the need to improve process energy efficiency have led to the widespread implementation of modern sulfur production units (SPUs) based on the Claus process [1].

Despite the high level of automation of modern industrial processes, sulfur production remains one of the most challenging control objects. This is due to the complex nonlinear relationship between input and output parameters, significant time delays, variability in feedstock composition, and the presence of various sources of uncertainty [2, 3]. In addition, sulfur quality is significantly affected by fluctuations in temperature, process pressure, air flow rate, hydrogen sulfide content, and other factors.

Conventional automatic control systems based on PID controllers and linear mathematical models are unable to account for uncertainty in process information or adapt in real time to changes in the characteristics of the controlled object. Consequently, they cannot provide effective control of such complex nonlinear processes, which leads to a lower sulfur recovery degree, increased sulfur dioxide concentration in tail gases, and deterioration in the quality of the sulfur produced.

In recent years, intelligent methods for controlling complex industrial processes based on artificial intelligence have developed significantly. Of particular interest are machine-learning methods, artificial neural networks (ANN), fuzzy logic (FL), genetic algorithms (GA), and their hybrid combinations [4–14]. Intelligent methods make it possible to describe complex nonlinear relationships without the need to construct exact physicochemical models of the process under stochastic and fuzzy uncertainty.

A special place among intelligent methods is occupied by the adaptive neuro-fuzzy inference system (ANFIS), which combines the advantages of artificial neural networks and fuzzy logic [15, 16]. Owing to its ability to learn automatically from experimental data, ANFIS provides high accuracy in approximating nonlinear relationships, robustness to measurement noise, and interpretability of the resulting control rules. However, ANFIS performance largely depends on the quality of the membership-function parameter tuning and the fuzzy rule base.

A genetic algorithm can be used to tune the membership-function parameters and the rule base effectively, enabling a global search for optimal parameters of the intelligent model while avoiding premature convergence to local extrema. The combined use of ANFIS and a genetic algorithm provides high prediction accuracy and high-quality control decisions for the high-quality sulfur production process under uncertainty.

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Sulfur production processes are characterized by uncertainties of various origins. The main sources of uncertainty include measurement-instrument errors, changes in feedstock composition, random fluctuations in process-stream flow rates, changes in environmental conditions, fuzziness of expert knowledge, and incomplete information about the internal state of the process object. Fuzzy logic is especially effective for accounting for such factors because it enables expert knowledge to be formalized and decisions to be made under insufficient certainty [17].

Therefore, a relevant scientific and practical task is the development of a hybrid intelligent control system (HICS) that integrates neuro-fuzzy modeling, genetic-algorithm-based optimization, and fuzzy inference for controlling the high-quality sulfur production process.

The aim of this study is to develop a HICS for the high-quality sulfur production process under uncertainty that improves sulfur quality, the sulfur recovery degree, and the environmental efficiency of the unit.

To achieve this aim, the following tasks are addressed:

– development of the HICS structure for the sulfur production process;

– formation of a set of informative process parameters and a fuzzy knowledge base;

– development of an ANFIS model for predicting product-quality indicators;

– development and application of a GA for optimizing the parameters of the intelligent system;

– conducting a computational experiment and analyzing the effectiveness of the proposed approach.

The scientific novelty of this work lies in the development of a hybrid intelligent control system that integrates an adaptive neuro-fuzzy inference system (ANFIS), a genetic algorithm, and fuzzy multi-criteria decision making to optimize the sulfur production process under uncertainty in process parameters.

The practical significance of the study is determined by the possibility of using the proposed intelligent system as part of automated process control systems at oil and gas processing enterprises to improve product quality, reduce harmful emissions, increase the sulfur recovery degree, and reduce operating costs.

II. Materials and Methods

The materials, data, and methods used in this study are as follows:

– data on process parameters and operating modes of the SPU, obtained and processed using experimental research methods [18] and mathematical statistics [19].

To develop the hybrid intelligent control system for the high-quality sulfur production process under uncertainty, the following intelligent methods integrated into a unified methodology are used:

– fuzzy-logic methods for creating and using fuzzy rules [2, 7–9, 20, 21];

– artificial neural network and machine-learning methods for predicting sulfur yield and quality and adapting the system to changing feedstock conditions and sulfur-production process parameters [4–6, 22, 23];

– expert-assessment methods for formalizing the expert knowledge of process operators and subject-matter experts [24, 25].

The object of the study is the SPU at the Atyrau refinery, designed to produce sulfur from harmful sulfur-containing gases released by other process units of the refinery [26].

Under actual operating conditions, the sulfur production process at this facility is affected by numerous uncertainty factors [27]:

– changes in feedstock composition and fluctuations in feed flow rate; changes in the temperature and pressure of SPU equipment;

– measurement-instrument errors;

– random external disturbances.

Therefore, the task of controlling the sulfur production process at the SPU belongs to the class of complex nonlinear multivariable systems with uncertain parameters.

Structure of the HICS. To improve the quality of sulfur-production process control, a HICS consisting of four interconnected modules was developed:

1. data preprocessing module;
2. ANFIS prediction module;
3. genetic optimization algorithm;
4. control-action generation module.

The HICS structure can be represented by the following sequence:

$$x \longrightarrow \text{Pre-processing} \longrightarrow \text{ANFIS} \longrightarrow \text{GA} \longrightarrow u^*,$$

where $x = [x_1, x_2, x_3, x_4, x_5, x_6]^T$ – is the vector of input parameters; u^*– is the optimal vector of control actions.

The following input variables are used: x_1– thermoreactor temperature, °C; x_2– air flow rate, m³/h; x_3– H_2S content, %; x_4– process pressure, MPa; x_5– excess air ratio; x_6– acid gas (feed) flow rate, m³/h.

The output vector is $y = [y_1, y_2, y_3]^T$, , where y_1– is sulfur yield; y_2– is the sulfur recovery degree; y_3– is SO_2 concentration.

ANFIS neuro-fuzzy model. An adaptive neuro-fuzzy inference system (ANFIS) is used to model the nonlinear relationship between the input parameters and sulfur yield and quality.

Mathematically, the model is written as:

$$y = f_{\text{ANFIS}}(\mathbf{x}).$$

Each input variable is described by three linguistic terms:

$$A_i = \{\text{Низкий}, \text{Средний}, \text{Высокий}\}.$$

To ensure a smooth response surface and a high learning rate, the membership functions are specified as Gaussian functions:

$$\mu(x) = \exp\left(-\frac{(x-c)^2}{2\sigma^2}\right),$$

where $\mu(x)$– is the membership degree; c– is the center of the membership function; σ – is the function-width parameter.

To form the fuzzy rule base, the Sugeno rule is used. Rule R_i,, where $i = \overline{1,729}$ is the rule number, has the form:–

$$R_i: IF\ x_1 = A_1^{(i)} \wedge x_2 = A_2^{(i)} \wedge \cdots \wedge x_6 = A_6^{(i)}, THEN$$
$$f_i = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{i6}x_6 + b_i.$$

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Here, a_{ij} – are the linear coefficients; b_i – is the bias term; these parameters are determined automatically during model training from experimental data.

Rule activation strength. The firing strength is determined by the product of the membership degrees

$$w_i = \prod_{j=1}^{6} \mu_{ij}(x_j).$$

Normalized activation strength

$$\overline{w}_i = \frac{w_i}{\sum_{k=1}^{N} w_k},$$

where N – is the number of rules.

ANFIS output. The final value is calculated as

$$y = \sum_{i=1}^{N} \overline{w}_i f_i.$$

When three terms are used for each of the six input variables, the number of rules is

$$N = 3^6 = 729.$$

Model training. The model parameters are adjusted using a hybrid learning algorithm that combines the least-squares method and the backpropagation algorithm.

Training objective function

$$J = \frac{1}{M}\sum_{i=1}^{M}(y_i - \hat{y}_i)^2,$$

where M is the number of training samples; $-y_i-$ is the experimental value of the output variable; $\hat{y}_i-$ is the value predicted by the model.

The mean squared error of the objective function is minimized [28]

$$RMSE = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(y_i - \hat{y}_i)^2}.$$

The membership-function parameters are updated according to the rule

$$\theta^{(t+1)} = \theta^{(t)} - \eta\frac{\partial J}{\partial \theta},$$

where $\theta^{(t)}$ is the vector of membership-function parameters at the $-t$-th training iteration, including all tunable parameters (c, σ); $\eta-$ is the learning rate; $\frac{\partial J}{\partial \theta}-$ is the gradient of the objective function J with respect to parameters θ, characterizing the direction and rate of change of the training error when the membership-function parameters are varied.

Genetic algorithm optimization. After the model is trained, the optimal operating mode of the SPU is searched for.

The chromosome has the form $z = [x_1, x_2, x_3, x_4, x_5, x_6]$.. Each gene represents one process variable. The fitness function is defined by the expression $F = \alpha_1 y_1 + \alpha_2 y_2 - \alpha_3 y_3$,, where $\alpha_1 + \alpha_2 + \alpha_3 = 1$ specify the relative importance of the optimization criteria.

Crossover operator. The offspring is generated according to the expression

$$z_i' = \lambda z_i^{(1)} + (1-\lambda) z_i^{(2)},$$

where $z_i'-$ is the value of the i-th gene of the offspring obtained by arithmetic crossover (the i-th optimized process parameter); λ is the crossover coefficient determining the contribution of each parent solution to the offspring; $-z_i^{(1)}$ и $z_i^{(2)}-$ is the value of the i-th gene (parameter) of the first and second parent solutions (chromosomes 1 and 2);

Mutation operator. Mutation is implemented according to the rule

$$z_i'' = z_i' + \delta,$$

where $z_i'-$ is the value of the i-th offspring parameter obtained after the crossover operation; $z_i''-$ is the value of the i-th parameter after the mutation operation; $\delta-$ is a random perturbation generated from a normal distribution with zero mean and variance σ^2: $\delta \sim N(0, \sigma^2)$.

Stopping criterion. The optimization process is terminated when condition [29] is satisfied:

$$| F^{(t+1)} - F^{(t)} | < \varepsilon,$$

where $F^{(t)}-$ is the fitness-function value at the current iteration; $F^{(t+1)}-$ is the fitness-function value at the next iteration; $\varepsilon-$ is the allowable tolerance.

III. Results and Discussions

Computational experiment. To evaluate the effectiveness of the proposed HICS, a computational experiment simulating the operation of the SPU under study was conducted. The experiment used a dataset formed from the process-parameter ranges specified by the operating conditions of the SPU at the Atyrau refinery.

The input variables were thermoreactor temperature $T = x_1$,, process air flow rate Q_{air}=x_2,, H_2S content H_2S in the feed (acid gas) $-C_{H_2S} = x_3$,, process pressure $P = x_4$,, excess air ratio $\lambda = x_5$, and acid gas flow rate $Q_{\text{gas}} = x_6$.. The output parameters were sulfur yield $Y_S = y_1$, sulfur recovery degree $\eta_S = y_2$, and sulfur dioxide concentration in the tail gases $C_{SO_2} = y_3$.

A total of 1000 observations were used to train the model; 75% were used for ANFIS training and 25% for testing. Before training, all features were normalized to the range $-[0; 1]$,, which ensured the stability of the optimization procedure and accelerated algorithm convergence.

The model was trained for 100 epochs using a hybrid learning algorithm combining the least-squares method and the backpropagation algorithm. The model parameters were optimized using the proposed genetic algorithm with a population size of 50 individuals, a crossover probability of 0.85, and a mutation probability of 0.05.

ANFIS model training results. During 100 epochs of training using the hybrid learning algorithm, the mean squared error (MSE) decreased steadily, demonstrating the model's strong ability to approximate nonlinear relationships between the input, operating, and output parameters of the SPU.

Figure 1 shows the dependence of MSE on the number of training epochs.

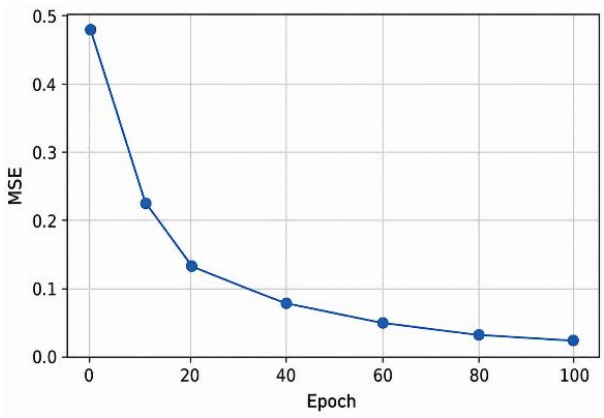


Figure 1. ANFIS learning curve

The resulting graph demonstrates a more than 17-fold decrease in training error, indicating good algorithm convergence and no signs of overfitting. The curve decreases monotonically and gradually reaches a plateau after the 80th epoch, indicating that the model has reached a stable state.

Prediction accuracy assessment. After training was completed, the ANFIS model was tested on an independent test set. Table 1 compares the actual sulfur-yield values obtained experimentally with the values predicted by the developed ANFIS neuro-fuzzy model.

Table 1 – Comparison of actual and predicted sulfur-yield values

№	Actual value, %	ANFIS prediction, %	Relative error, %
1	97,20	97,08	0,12
2	97.65	97.81	0.16
3	98.10	98.02	0.08
4	98.45	98.34	0.11
5	98.83	98.94	0.11
6	99.05	98.96	0.09
7	99.22	99.14	0.08
8	99.38	99.47	0.09

The mean relative prediction error was 0.11%, confirming the high accuracy of the developed ANFIS neuro-fuzzy model.

Figure 2 presents the "Actual Values – ANFIS Prediction" scatter plot.

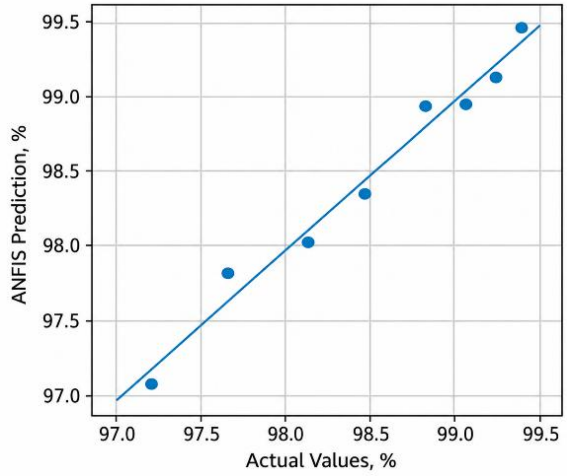


Figure 2. Scatter plot "Actual Values (x) – ANFIS Prediction (y)".

As can be seen from Figure 2, most points lie close to the line $y = x$, indicating high prediction accuracy.

For a comprehensive assessment of prediction quality, the standard performance metrics were calculated: root mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and the coefficient of determination (R^2).. The model-quality assessment results for these metrics are presented in Table 2.

Table 2 – Model performance metrics

Metric	Value
RMSE	0.029
MAE	0.021
MAPE	0.23 %
R^2	0.992

The obtained values show that the model provides an almost complete agreement between the calculated and experimental data.

Optimization of the high-quality sulfur production process. After the developed ANFIS model had been trained, multiobjective optimization of the sulfur production process was performed using the genetic algorithm proposed in Section II.

Optimization was performed simultaneously according to three criteria: maximizing sulfur yield y_1;, maximizing the sulfur recovery degree y_2; , and minimizing SO_2 concentration in the tail gases y_3. The optimization results based on the proposed genetic algorithm are presented in Table 3.

Table 3 – Optimization results

Parameter	Before optimization	After optimization
Sulfur yield, %	97,8	99,1
Sulfur recovery degree, %	98,3	99,6
SO_2 concentration, ppm	295	188

The obtained optimization results show that application of the hybrid intelligent system made it possible to:

– increase sulfur yield by 1.3%;

– increase the sulfur recovery degree by 1.3%;

– reduce sulfur dioxide concentration in the tail gases by more than 36%.

The optimization results are technologically significant because reducing sulfur dioxide (SO_2) emissions directly contributes to compliance with environmental standards and reduces the load on downstream gas-cleaning systems.

Pareto front analysis. Multiobjective optimization using the proposed genetic algorithm produced a set of Pareto-optimal solutions characterizing the trade-off between sulfur yield and environmental performance indicators [30]. Figure 3 illustrates the Pareto front obtained by multiobjective optimization using the proposed genetic algorithm.

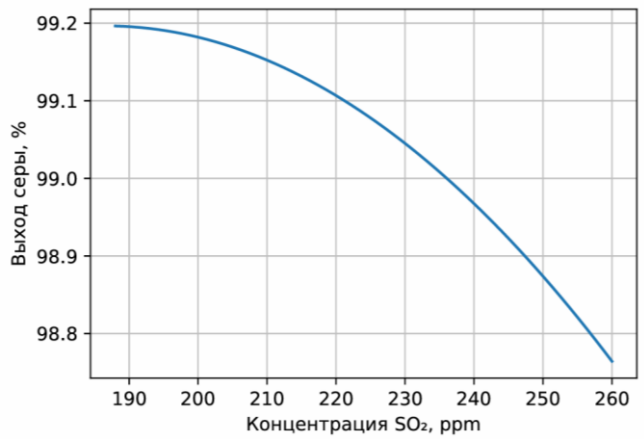


Figure 3. Pareto front obtained by multiobjective optimization based on the genetic algorithm

This figure illustrates the Pareto front, where each point corresponds to one feasible operating mode of the SPU under study at the Atyrau refinery.

Analysis of the Pareto front shows that a further increase in sulfur yield above 99.2% is accompanied by higher energy consumption and a slight increase in SO_2 concentration. Therefore, the most rational operating modes are those located in the central part of the Pareto front, providing an optimal balance between the economic and environmental efficiency of production.

Comparison of the proposed HICS with other known control methods. To evaluate the effectiveness of the proposed hybrid intelligent control system, it was compared with conventional PID control and a standalone ANFIS model without genetic optimization. The comparison results are presented in Table 4.

Table 4 – Comparison of control methods

Control method	RMSE	Sulfur yield, %	SO_2 concentration, ppm
PID controller	0,094	97,2	326
ANFIS	0,041	98,5	241
Proposed hybrid system (ANFIS + GA)	**0,029**	**99,1**	**188**

The comparison results show that integrating neuro-fuzzy modeling with a genetic algorithm significantly improves control accuracy and process quality compared with both conventional PID control and the use of the ANFIS model alone.

Discussion of results. The obtained results confirm the effectiveness of the proposed hybrid intelligent control system for the high-quality sulfur production process under uncertainty. ANFIS adequately described the complex nonlinear fuzzy relationships between the input and output parameters of the SPU, while the genetic algorithm provided global optimization of the unit operating modes.

Of particular importance is the model's high robustness to uncertainty in the input information. When the level of random disturbances was increased to 15%, the deterioration in prediction accuracy was insignificant, indicating the possibility of practical application of the developed system under actual industrial conditions, where measurement noise and changes in feedstock composition are inevitable.

Multiobjective optimization using the proposed genetic algorithm produced a set of Pareto-optimal solutions (Figure 3). Analysis of the resulting Pareto front shows that a further increase in sulfur yield above 99.2% is accompanied by higher energy consumption and a slight increase in sulfur dioxide concentration in the tail gases. This indicates that the most rational SPU operating modes are those located in the central part of the Pareto front, providing an optimal balance between the economic and environmental efficiency of the unit.

Compared with conventional automatic-control methods, the proposed approach provides higher prediction accuracy, increased sulfur yield, reduced sulfur dioxide emissions, and improved overall efficiency of the high-quality sulfur production process. This creates the prerequisites for integrating the developed hybrid intelligent system into modern automated process control systems (APCS) for sulfur production and other products at oil and gas processing enterprises.

IV. Conclusions

In this study, a hybrid intelligent control system for the high-quality sulfur production process under uncertainty in process information was developed. The proposed approach is based on integrating an adaptive neuro-fuzzy inference system (ANFIS), fuzzy logic, and a genetic algorithm for multiobjective optimization, which makes it possible to account for fuzziness, process nonlinearity, uncertainty in the input data, and the multi-criteria nature of control.

A mathematical model of the HICS was developed, including a process-data preprocessing module, a neuro-fuzzy prediction block, a genetic optimization module, and a system for generating optimal control actions. The main process variables are thermoreactor temperature, air flow rate, hydrogen sulfide content, process pressure, excess air ratio, and feed flow rate.

The computational experiment confirmed the operability of the proposed HICS. The results demonstrated high accuracy in predicting process indicators and a steady decrease in the ANFIS model training error.

The proposed genetic algorithm made it possible to determine rational operating modes of the sulfur production unit and to form a set of Pareto-optimal solutions providing a trade-off between maximum sulfur yield, a high sulfur recovery degree, and minimization of sulfur dioxide emissions. Compared with known control methods, the proposed hybrid control system improves the quality of the sulfur production process and reduces the environmental burden.

The scientific novelty of the study lies in the development of a hybrid intelligent control system for the high-quality sulfur production process that integrates an adaptive neuro-fuzzy inference system (ANFIS), a genetic algorithm, and fuzzy multi-criteria decision making to optimize the sulfur production process under uncertainty.

The practical value of the study lies in the possibility of integrating the developed system into modern automated process control systems at oil and gas processing enterprises. The proposed approach can be used for the digital modernization of sulfur production units, implementation of

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the digital-twin concept for the process object, and development of intelligent decision-support systems.

Promising directions for further research include developing a digital twin of the SPU, applying deep-learning and reinforcement-learning methods for adaptive real-time control, and integrating the hybrid intelligent system with industrial SCADA systems.

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